

OGUNBIYI ELIJAH

AI & Backend Engineer | Agentic Systems · Go · TypeScript · Distributed Systems

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SUMMARY

AI and backend engineer with four years of production experience, mostly in Go and TypeScript, with Python, Kotlin and C where the problem calls for them. At Lustrew Dynamics I build the execution engine behind AEGIS ADA, a commercial AI operator that reads a user's screen and drives their applications for them, along with the resolution layer that lets it work identically across Win32 desktop applications, browsers over the Chrome DevTools Protocol, and Microsoft Office.

Most of what I build is shaped by one habit: making correctness structural instead of a matter of discipline. If a boundary matters, a linter enforces it in CI. If the same pricing logic lives in two runtimes, a shared fixture suite proves the two agree. If an agent can take a destructive action, the permission model makes that action previewable before it happens rather than confirmable after. I have shipped web platforms, Android apps and backend services, and can own any of them end to end.

PROFESSIONAL EXPERIENCE

Lustrew Dynamics, Ikeja, Lagos, Nigeria (On-site) · lustrewdynamics.com July 2025 – Present

AI Engineer January 2026 – Present

- Build the execution engine behind AEGIS ADA (aegisada.com), Lustrew's commercial AI operator: multi-step planning, tool orchestration, and decisions made at runtime rather than scripted in advance. ADA ships as a desktop Copilot on Windows, macOS and Linux, as iOS and Android apps, and as a Chrome extension, on paid tiers up to enterprise.
- Designed the multimodal pipeline ADA perceives through. Screen content, documents, images and video arrive on one path, so the same reasoning applies whether it is reading a rendered interface or a spreadsheet.
- Built the cross-platform element resolution layer covering native Windows (Win32), browsers via the Chrome DevTools Protocol, and Microsoft Office. Three very different surfaces, one interaction model.
- Set the agent's tool boundaries and approval model, so every action is previewed and confirmed before it counts, and a step that fails partway degrades cleanly instead of leaving a task half-done.

Full-Stack Software Engineer July 2025 – December 2025

- Built and shipped features across the stack, from data models and API endpoints through to the interfaces consuming them.
- Moved onto the agent work after six months as the company began building ADA out, and have led that side of the engineering since.

OTESC (Remote), Software Engineer 2023 – Present

- Design and maintain the REST services and the Next.js clients on top of them, owning the schema, the API contract, JWT authentication and the deploy.
- Led the move off the legacy stack onto FastAPI and Next.js, which mostly meant replacing undocumented endpoints with versioned, specified ones.
- Set up CI/CD on Docker and GitHub Actions so releases stopped being a manual step somebody had to remember.

Food Surplus Limited (Remote), Backend Engineer, Contract 2025, six months

- Built the inventory management system for multi-site food distribution on FastAPI and MySQL.
- Wrote the APIs behind real-time stock tracking across five distribution centres, including the consistency handling for concurrent updates to the same item.
- Added role-based access control separating admin, operations and distribution-centre permissions.

Bincom Dev Center, Lagos, Nigeria, Mobile App Developer Intern 2022 – 2023

- Built cross-platform features in React Native that shipped to both iOS and Android.
- Profiled and refactored the slow parts of the render path, and tracked down the platform-specific crashes that are most of what mobile debugging turns out to be.

SELECTED PROJECTS

LettsTV In production · lettstv.com · Next.js, React, Node.js

- A live news and media platform: article publishing, an editorial admin, topic pages, social login, inline PDF rendering, and the SEO and OpenGraph handling a publisher actually needs.
- Sole engineer across 17 months, from the first commit through to live traffic and the feature work that follows it.

OTESC Store In development · Go, Next.js, PostgreSQL, Redis, Expo

- Seven applications (customer, admin and seller web, plus Expo apps for shop, seller and rider) on one layered Go modular monolith, arranged as a Go workspace alongside a pnpm and Turborepo monorepo.

- The architecture is enforced by CI rather than by convention. go-arch-lint holds the transport, service, repository and domain layers in the right dependency direction, and dependency-cruiser with eslint-plugin-boundaries does the same for the TypeScript packages, so UI code cannot reach for the API client.
- Pricing, the order state machine and the geo maths exist in both Go (authoritative) and TypeScript (preview). A shared set of golden fixtures is asserted by both test suites, so the two implementations cannot quietly drift apart.
- Split shipping was the hardest piece. One checkout can deliver to several addresses, each its own order with its own rider and tracking but sharing a single Paystack charge, and refunds are capped to the group's remaining balance so refunding siblings can never exceed what was charged.
- Geocoding is keyless by default (Photon over OpenStreetMap, OpenRouteService routing, VROOM batching), with Google Maps an opt-in upgrade behind one environment variable, keeping a per-request cost off the critical path.

JARVIS / Freya Personal project · Go, standard library only

- A local-first AI assistant with an empty go.mod. No external dependencies at all, so it builds offline in seconds and ships as a single static binary.
- The permission model has three tiers rather than two, because the real risk is not a malicious user but a capable model acting on an ambiguous instruction. Actions either run silently, or run only after showing a concrete preview of what will change, or do not run at all regardless of approval. A plain yes/no prompt only teaches people to say yes.
- Memory is tiered and ordered most stable first, so the model's prompt cache keeps hitting instead of the full context being resent every turn.

Rooster Play Store submission pending · Kotlin (Android), Go

- An Android messenger on a Go backend: channels, media and video messaging, full account lifecycle, and accessibility and data-saving settings that actually do something.
- Contact discovery runs on hashed matching, so the server never receives a plaintext address book.

Chomp Play Store submission pending · C, raylib, Go

- Written in C with raylib: procedurally generated mazes, each validated by flood fill, and four enemy behaviours running on scatter and chase waves.
- Two-player co-op in real time over a Go relay. The host runs the simulation while the guest predicts its own movement and reconciles, so neither player feels input lag, and ninety seconds of grace lets a dropped player retake their seat.

Postal In development · Go, PostgreSQL, Redis

- A social publishing backend. Channels connect over OAuth 2.0 PKCE with platform tokens envelope-encrypted at rest (AES-256-GCM), publishing is idempotent so retries never double-post, scheduling runs on asynq, and the API surface is specified in a hand-written OpenAPI 3.0 document.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, Kotlin, C, SQL

AI & Agentic Systems: Autonomous agent design, execution engines, tool orchestration, multimodal pipelines, agent permission and safety models, LLM integration (Gemini, Claude), cross-platform UI automation (Win32, DOM/CDP, Microsoft Office)

Backend: Go (chi, pgx, sqlc, asynq), FastAPI, Django REST Framework, Node.js, Express, Bun, Elysia

Frontend & Mobile: Next.js, React, React Native, Expo, Tailwind CSS, TanStack Query, Redux

Data: PostgreSQL, MySQL, MongoDB, Redis, S3-compatible object storage (Cloudflare R2, MinIO)

Security: OAuth 2.0 PKCE, JWT with rotating refresh sessions, Argon2id, bcrypt, AES-256-GCM encryption at rest, HMAC, capability-based and role-based authorization, rate limiting, audit logging

Reliability & Architecture: Idempotent job pipelines, queue-based scheduling, retry and backoff error models, multi-tenant isolation, layered modular monoliths, architecture linting in CI (go-arch-lint, dependency-cruiser), cross-runtime parity testing, property and invariant testing

Infrastructure: Docker, Kubernetes, Helm, GitHub Actions, Caddy, goose migrations, Turborepo, pnpm, OpenAPI 3.0, CI/CD

EDUCATION

University of the People. B.Sc. Computer Science, in progress, expected 2026. Associate of Science in Computer Science, completed.

Federal University of Technology, Akure (FUTA), Nigeria. B.Sc. Information and Communication Technology. 2018 to 2023.

CERTIFICATIONS

Oracle APEX and Database Management. Oracle University.